D222 local/global grouped-query decoder

Task and source

The Phase 1 task is 128-word legal continuation given a 512-word prompt. The four official sources are Acts, High Court, Supreme Court, and Tribunal material. This report is a later reconstruction based on recovered source and archived run evidence, not an executed training notebook.

Architecture

A 24-block, width-896 causal decoder uses 14 query heads over two shared key/value heads. Every fourth block attends globally and intervening blocks use 512-token local attention. The 222M-scale run was built after the original Phase 1 deadline.

Stage	512-word prompt -> Own byte BPE -> 24 local/global blocks
Continuation	Grouped-query attention -> SwiGLU / tied head -> 128-word continuation

Token and optimization science

The model learns by minimizing next-token cross-entropy under causal masking. An attention head compares query/key vectors, obtains a weighted value mixture, and updates the residual stream; rotary phases give position-sensitive comparisons. Decoding then turns logits into a 128-word continuation. Tokenizer choice affects sequence length, vocabulary utilization, and the optimization budget. The original source config and scripts identify the actual optimizer, schedule, context and generation policy.

Reported outcome

Local confirmation BERTScore F1 0.863190; post-deadline, no Codabench score. Historical screenshot numbers are not freshly retrieved leaderboard results. Archived local metric files, when present, provide separately measured BERTScore, ROUGE-L and BLEU-4; no metric is invented for runs without such evidence.

Reproduction boundary

The code package contains the recovered model, tokenizer, training, evaluation and export scripts, plus original small logs/metrics. The complete ZIP holds selected checkpoint weights and official data, which are too large or inappropriate to duplicate in GitHub. Re-execution requires their original hashes and path remapping.

Configured dimensions

The following values are read directly from the original configuration. A continued-training config may inherit the parent model graph; that relationship is identified in the README and source code.

Setting	Archived value
vocab_size	32768
layers	24
width	896
query_heads	14
kv_heads	2
head_dim	64
ff_width	2304
context	2048
local_window	512
global_every	4
query_tile	128
dropout	0.0
checkpointing	True

Archived evaluations

This table is a small sample of per-policy and per-checkpoint records, not a single comparable leaderboard ranking. Full machine-readable values and names are in evidence/archived_metric_index.json.

Metric file	BERTScore F1	ROUGE-L F1	BLEU-4
benchmark/phase1_benchmark/candidate_0019250_...	0.8498	0.3339	0.2311
benchmark/phase1_benchmark/candidate_0019606_...	0.8626	0.3262	0.2367
benchmark/phase1_benchmark/confirmation.metri...	0.8632	0.3303	0.2485
benchmark/phase1_benchmark/candidate_0019250_...	0.8594	0.3142	0.2220
benchmark/phase1_benchmark/candidate_0019606_...	0.8512	0.3477	0.2530
benchmark/phase1_benchmark/candidate_0019500_...	0.8638	0.3247	0.2371
benchmark/phase1_benchmark/candidate_0019500_...	0.8479	0.3245	0.2278

Evidence count and caveat

Copied 10 compact log/metric/manifest artifacts from CS781_D222_Benchmark_Complete_20261012.zip. They support the run lineage and measured local evaluations. A lower generation loss need not imply better BERTScore or legal accuracy; checkpoint and decoding policies must be evaluated separately. Original October 6 source archive contains no .ipynb file, so a later illustrated notebook is explicitly marked as reconstructed.